Chang Jin

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Education

Tongji University *Shanghai, China*
Master of Computer Science *Sep 2023 - Present*

- GPA: **4.84**/5.00

École Polytechnique Fédérale de Lausanne (EPFL) *Lausanne, Switzerland*
Exchange Master Program of Computer Science *Feb 2025 - Jul 2025*

- GPA: **5.62**/6.00

Tongji University *Shanghai, China*
Bachelor of Computer Science *Sep 2018 – Jun 2023*

- GPA: **4.91**/5.00
- Rank: **3**/121

Publications

When Silence Is Golden: Can LLMs Learn to Abstain in Temporal QA and Beyond?
Xinyu Zhou, **Chang Jin** (co-first author), Carsten Eickhoff, Zhijiang Guo, Seyed Ali Bahrainian
Submitted to *ICLR 2026*

Effective and Explainable Molecular Property Prediction by a Chain-of-Thought Enabled LLM and Multi modal Molecular Information Fusion
Chang Jin, Siyuan Guo, Shuigeng Zhou, Jihong Guan
Published in *Journal of Chemical Information and Modeling* (JCR Q1)

M3-20M: A Large-Scale Multi-Modal Molecule Dataset for AI-driven Drug Design and Discovery
Siyuan Guo, Lexuan Wang, **Chang Jin**, Jinxian Wang, Han Peng, Huayang Shi, Wengen Li, Jihong Guan, Shuigeng Zhou
Published in *Journal of Bioinformatics and Computational Biology*

Enhanced Adaptive Graph Convolutional Network for Long-Term Fine-Grained SST Prediction
Han Peng, **Chang Jin** (co-first author), Wengen Li, Jihong Guan
Published in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (JCR Q1)

On Evaluating the Predictability of Sea Surface Temperature Using Entropy
Chang Jin, Han Peng, Hanchen Yang, Wengen Li, Jihong Guan
Published in *Remote Sensing* (JCR Q1)

Research Experience

Research on Chain-of-Thought Monitoring for Large Language Models *Sep 2025 – Present*
Advisor: Dr. Xingcheng Xu, Dr. Qiaosheng Zhang (Shanghai Artificial Intelligence Laboratory)

- Conduct research on chain-of-thought (CoT) monitoring for large language models, aiming to improve the faithfulness and safety of reasoning processes.
- Explore methods to detect and mitigate obfuscated or adversarial reasoning traces, and develop monitoring frameworks for reliable model behavior under different optimization pressures.

Research on Enhancing Abstention Mechanisms of Large Language Models for Temporal Reasoning Tasks *Sep 2024 – Present*
Advisor: Dr. Ali Bahrainian, Prof. Carsten Eickhoff (Joint Lab, University of Tübingen & Brown University)

- Developed reinforcement learning based techniques to enhance LLM understanding of temporal knowledge.

- Improved model abstention mechanisms, enabling LLMs to abstain from answering unanswerable queries, reducing hallucinations and errors.
- **Research Output**: Co-first author: a manuscript submitted to ICLR 2026.

Research on Constrained Decoding Methods for Structured Text Generation in Large Language Models *Feb 2025 – Present*
Advisor: Prof. Robert West (EPFL)

- Investigated capabilities of various LLMs to generate JSON objects that adhere to given JSON Schemas.
- Analyzed probability distributions during constrained decoding to reveal how schema constraints affect LLM generation behavior.
- **Research Output**: Contributed a JSON Schema–based evaluation task to the open-source project lm-evaluation-harness (9.5k stars), enabling structured-output LLM evaluation.

Research on Drug Molecule Discovery Using Large Language Models *Sep 2023 – Sep 2025*
Advisor: Prof. Jihong Guan (Tongji University) & Prof. Shuigeng Zhou (Fudan University)

- Constructed M3-20M, a large-scale multi-modal molecular dataset with over 20 million molecules.
- Developed LLM-MPP, a Chain-of-Thought enhanced multimodal LLM for molecular property prediction, achieving state-of-the-art performance in prediction accuracy and interpretability.
- **Publications**:
  - First author: Effective and Explainable Molecular Property Prediction by a Chain-of-Thought Enabled LLM and Multimodal Molecular Information Fusion, *Journal of Chemical Information and Modeling (JCR Q1)*.
  - Co-author: M3-20M: A Large-Scale Multi-Modal Molecule Dataset for AI-driven Drug Design and Discovery, *Journal of Bioinformatics and Computational Biology*.

Research on Spatio-temporal Modeling *Mar 2021 – Jun 2023*
Advisor: Prof. Jihong Guan & Prof. Wengen Li (Tongji University)

- Developed EA-GCN, a spatio-temporal deep learning model for long-term fine-grained sea surface temperature (SST) prediction, achieving state-of-the-art performance.
- Proposed a temporal-correlated entropy method to evaluate SST predictability from global and local perspectives, aiding marine and climate monitoring.
- **Publications**:
  - Co-first author: Enhanced Adaptive Graph Convolutional Network for Long-Term Fine-Grained SST Prediction, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JCR Q1)*.
  - First author: On Evaluating the Predictability of Sea Surface Temperature Using Entropy, *Remote Sensing (JCR Q1)*.

Honors and Awards

Outstanding Graduate of Tongji University	Jun 2023
First-Class Scholarship of Tongji University (top 5%)	2019, 2021
Second Prize of China Undergraduate Mathematical Contest in Modeling, Shanghai	2020

Skills

Programming Languages: Python, C++, C

Python Packages: PyTorch, TensorFlow, PyG, sklearn and other packages related to deep learning.

English Skills: IELTS: 7.5 (Listening: 7.5, Reading: 8.5, Writing: 7.5, Speaking: 6.5)